Selective Tests of Penalized-Fit Shapley Contributions after Cross-Validation

2 Problem Setup

2.1 Statistical model and fixed auxiliary information

Consider a regression dataset $(X, \boldsymbol{y})$ with n observations and m candidate features, where $X = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m] \in \mathbb{R}^{n \times m}$ is fixed, X_{ij} is its (i, j) entry, and $\boldsymbol{y} \in \mathbb{R}^n$ is a realization of

$$\boldsymbol{Y} = \boldsymbol{\mu}(X) + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n), \qquad \sigma > 0 \text{ known.} \tag{1}$$

Here I_n is the $n \times n$ identity matrix and $\mathbf{0}$ denotes a zero vector of the required dimension. Write $\boldsymbol{\mu} = \boldsymbol{\mu}(X)$ and denote the observed response by $\boldsymbol{y}_{\text{obs}}$; $\boldsymbol{y}$ denotes a generic response vector. We impose no parametric representation of $\boldsymbol{\mu}$: the regression basis below defines the contribution being tested, rather than a requirement that the true mean belong to that basis. All probability statements condition on X. They also condition on auxiliary random choices, denoted collectively by ω, that determine basis centers and cross-validation folds. These choices are independent of the response noise given X. The basis fraction, candidate penalty grid, and number k of selected features are specified before inspecting $\boldsymbol{Y}$.

2.2 Coalition models and response-dependent tuning

Let $[m] = \{1, \ldots, m\}$, and fix $k \in \{1, \ldots, m\}$. For a prespecified fraction $\rho \in (0, 1]$, sample $m_{\text{RBF}} = \max\{1, \lceil \rho n \rceil\}$ distinct row indices $i_1, \ldots, i_{m_{\text{RBF}}}$ using ω, sharing these indices across features. For feature $j \in [m]$, define centers $c_{jq} = X_{i_q j}$ and, for $q = 1, \ldots, m_{\text{RBF}}$, $i = 1, \ldots, n$, and $t \in \mathbb{R}$, set

$$g_{jq}(t) = \exp\left[-\frac{(t - c_{jq})^2}{2s_j^2}\right], \qquad (G_j)_{iq} = g_{jq}(X_{ij}) - \frac{1}{n}\sum_{a=1}^{n} g_{jq}(X_{aj}). \tag{2}$$

The width s_j is the median of $|X_{aj} - X_{bj}|$ over $1 \le a < b \le n$. If this is zero, use the median strictly positive distance; if there are no positive distances, use $s_j = 1$. Let $\mathbf{1}_n$ denote the vector of n ones. Thus $G_j \in \mathbb{R}^{n \times m_{\text{RBF}}}$ is determined by X and ω, and $\mathbf{1}_n^\top G_j = \mathbf{0}^\top$. For $S \subseteq [m]$, let $G_S \in \mathbb{R}^{n \times m_{\text{RBF}}|S|}$ concatenate the blocks indexed by S, with $G_\varnothing$ having zero columns. These are additive univariate Gaussian basis models; the construction contains no multivariate interaction basis.

For a fixed $\lambda > 0$, define the minimized penalized training objective

$$L_\lambda(S; \boldsymbol{y}) = \min_{\beta_0, \boldsymbol{\beta}_{-0}} \left\{ \|\boldsymbol{y} - \beta_0 \mathbf{1}_n - G_S \boldsymbol{\beta}_{-0}\|_2^2 + n\lambda \|\boldsymbol{\beta}_{-0}\|_2^2 \right\}. \tag{3}$$

Here $\beta_0 \in \mathbb{R}$ is the unpenalized intercept, $\boldsymbol{\beta}_{-0} \in \mathbb{R}^{m_{\mathrm{RBF}}|S|}$ contains the basis coefficients. With $P_0 = \mathbf{1}_n \mathbf{1}_n^\top / n$, $C_0 = I_n - P_0$, and $K_S = G_S G_S^\top$, set

$$H_S^\lambda = K_S (K_S + n\lambda I_n)^{-1}, \qquad H_\varnothing^\lambda = 0.$$

For $j \in [m] \setminus S$, centering gives

$$\begin{aligned} L_\lambda(S; \boldsymbol{y}) &= \boldsymbol{y}^\top (C_0 - H_S^\lambda) \boldsymbol{y}, \\ \Delta_j^\lambda(S; \boldsymbol{y}) &= L_\lambda(S; \boldsymbol{y}) - L_\lambda(S \cup \{j\}; \boldsymbol{y}) \\ &= \boldsymbol{y}^\top (H_{S \cup \{j\}}^\lambda - H_S^\lambda) \boldsymbol{y}. \end{aligned} \tag{4}$$

The increment matrix is positive semidefinite. Indeed, adding a block increases K_S in the Loewner order and $H_S^\lambda = I_n - n\lambda(K_S + n\lambda I_n)^{-1}$. Consequently $\Delta_j^\lambda(S; \boldsymbol{y}) \geq 0$. Equation (4) concerns the *minimized penalized objective*, including the ridge penalty. Replacing it with the residual sum of squares evaluated at a ridge fit would define a different score.

We choose λ by five-fold cross-validation on the full-feature model over the ordered grid

$$\Lambda = (\lambda_1, \ldots, \lambda_9) = (10^{-6}, 10^{-5}, \ldots, 10^2).$$

Within each fold, centers, widths, and centering means are constructed from the training rows alone, and the penalty in (3) uses that fold's training sample size. With X and ω fixed, each validation prediction is linear in $\boldsymbol{y}$. Therefore there are fixed symmetric positive semidefinite matrices C_ℓ such that the pooled validation mean squared error is

$$\mathrm{CV}_\ell(\boldsymbol{y}) = \boldsymbol{y}^\top C_\ell \boldsymbol{y}, \qquad \widehat{\ell}(\boldsymbol{y}) = \operatorname*{argmin}_{\ell \in \{1, \ldots, 9\}} \boldsymbol{y}^\top C_\ell \boldsymbol{y}, \qquad \widehat{\lambda}(\boldsymbol{y}) = \lambda_{\widehat{\ell}(\boldsymbol{y})}. \tag{5}$$

Ties are resolved toward the smaller penalty. The full-data basis is then used with the same selected penalty for every coalition. There is no coalition-specific tuning.

2.3 Shapley ranking and the selected target

For a nonempty feature universe $U \subseteq [m]$ and $j \in U$, define

$$w_{j,U}(S) = \frac{|S|!(|U| - |S| - 1)!}{|U|!} = \frac{1}{|U| \binom{|U|-1}{|S|}}, \qquad S \subseteq U \setminus \{j\},$$

$$A_{j,U}^\lambda = \sum_{S \subseteq U \setminus \{j\}} w_{j,U}(S) (H_{S \cup \{j\}}^\lambda - H_S^\lambda), \qquad T_{j,U}^\lambda(\boldsymbol{y}) = \boldsymbol{y}^\top A_{j,U}^\lambda \boldsymbol{y}. \tag{6}$$

Thus $T^{\lambda}_{j,U}$ averages the improvement from adding feature j over all insertion orders within U. The matrices are positive semidefinite, and the allocation satisfies

$$\sum_{j\in U} T^{\lambda}_{j,U}(\boldsymbol{y}) = L_\lambda(\varnothing;\boldsymbol{y}) - L_\lambda(U;\boldsymbol{y}). \tag{7}$$

The coalition sums in (6) are evaluated exhaustively. Rank the features by decreasing $T^{\widehat{\lambda}(\boldsymbol{y})}_{j,[m]}(\boldsymbol{y})$, breaking ties by increasing feature index, and denote the unordered top-k set by $M(\boldsymbol{y})$. Independently generate $V \sim \mathrm{Unif}[0,1)$ and set

$$J(\boldsymbol{y}, V) = M(\boldsymbol{y})_{(1+\lfloor kV \rfloor)}, \tag{8}$$

where the elements of $M(\boldsymbol{y})$ are listed in increasing index order. Conditional on $M(\boldsymbol{y})$, this chooses one target uniformly. The realized value v is held fixed during inference.

Let $\ell_0 = \widehat{\ell}(\boldsymbol{y}_{\mathrm{obs}})$, $\lambda_0 = \lambda_{\ell_0}$, $M_0 = M(\boldsymbol{y}_{\mathrm{obs}})$, and $j = J(\boldsymbol{y}_{\mathrm{obs}}, v)$. We consider two distinct targets for this same selected feature:

$$\begin{aligned} &U = M_0 \quad \text{(Selected-Features Shapley contribution)},\\ &U = [m] \quad \text{(All-Features Shapley contribution)}. \end{aligned} \tag{9}$$

Both use the ranking based on the full feature universe. The Selected-Features target does not rerank features using a restricted-universe score. The two procedures share the CV winner, selected set, and target feature.

2.4 Hypotheses and their interpretation

For the realized U, j, ℓ_0, abbreviate $A = A^{\lambda_0}_{j,U}$ and define the mean-response contribution

$$T^{\lambda_0}_{j,U}(\boldsymbol{\mu}) = \boldsymbol{\mu}^\top A \boldsymbol{\mu}.$$

The ideal zero-contribution hypotheses are

$$H_{0,j,U} : T^{\lambda_0}_{j,U}(\boldsymbol{\mu}) = 0 \quad \text{versus} \quad H_{1,j,U} : T^{\lambda_0}_{j,U}(\boldsymbol{\mu}) > 0. \tag{10}$$

These parameters refer to applying the coalition objective to the noise-free mean vector. They are not the expectation of the observed attribution: for fixed A, $\mathbb{E}_{\boldsymbol{\mu}}[\boldsymbol{Y}^\top A \boldsymbol{Y}] = \boldsymbol{\mu}^\top A \boldsymbol{\mu} + \sigma^2 \operatorname{tr}(A)$.

Let $W_j = \mathrm{range}(A)$, P_{W_j} be its orthogonal projector, and $d_j = \dim W_j$; the dependence of W_j and d_j on U and λ_0 is suppressed. Since $A \succeq 0$,

$$\boldsymbol{\mu}^\top A \boldsymbol{\mu} = 0 \quad \Longleftrightarrow \quad P_{W_j}\boldsymbol{\mu} = \mathbf{0}. \tag{11}$$

Moreover, positivity of every Shapley weight implies that the null in (10) holds exactly when $\Delta^{\lambda_0}_j(S;\boldsymbol{\mu}) = 0$ for every $S \subseteq U \setminus \{j\}$. Hence the test concerns the absence of contribution across *all*

coalitions in the chosen universe. It does not isolate a coefficient in the full additive model or assert conditional independence or causality.

The feature label need not imply a small test space. Writing $\mathcal{V}_U = \mathrm{col}(G_U)$, one has $W_j \subseteq \mathcal{V}_U$. If $W_j = \mathcal{V}_U$, (10) becomes the absence of mean signal in that entire centered basis space. Different features can have different attribution scores but the same zero-contribution null.

Numerical specification of the test space. For numerical computation, we first compress $[G_1, \ldots, G_m]$ by retaining singular values exceeding $\tau \max(1, s_{\max})$, with $\tau = 10^{-10}$, where $s_{\max}$ is the largest singular value. Coalition matrices are constructed in this retained space. From the resulting target matrix, we retain eigenvectors whose eigenvalues exceed $\tau \max(1, a_{\max})$. Here $a_{\max}$ is the largest eigenvalue of the compressed target matrix. If Q contains the retained orthonormal left singular vectors of the full basis and B contains these orthonormal target eigenvectors in Q coordinates, the test uses

$$W_{j,\tau} = \mathrm{col}(QB), \qquad H^{(\tau)}_{0,j,U} : P_{W_{j,\tau}} \boldsymbol{\mu} = \mathbf{0}. \tag{12}$$

All subsequent formulas apply with this numerically specified $W_{j,\tau}$ as well as with ideal W_j. Equivalence to the ideal quadratic null requires agreement of the retained and ideal ranges; a discarded small positive eigenvalue is not an exact mathematical zero. The tolerance is therefore part of the target specification.

2.5 Conditional inferential procedure

For either specified test space, let P denote its orthogonal projector (P_{W_j} or $P_{W_{j,\tau}}$) and $d_j = \mathrm{rank}(P)$ its rank. When $d_j \geq 1$, define

$$R_j(\boldsymbol{Y}) = \frac{\|P\boldsymbol{Y}\|_2}{\sigma}, \qquad \mathcal{Q}_{\boldsymbol{Y}} = (I_n - P)\boldsymbol{Y}, \qquad \boldsymbol{D}_j(\boldsymbol{Y}) = \frac{P\boldsymbol{Y}}{\|P\boldsymbol{Y}\|_2}. \tag{13}$$

For fixed P, under $P\boldsymbol{\mu} = \mathbf{0}$, the radius has distribution χ_{d_j} and is independent of the direction $\boldsymbol{D}_j(\boldsymbol{Y})$ and orthogonal component $\mathcal{Q}_{\boldsymbol{Y}}$. Conditional on the realized CV winner, selected set, and v, the projector is fixed. For $\boldsymbol{z}_j = \mathcal{Q}_{\boldsymbol{y}_{\mathrm{obs}}}$ and $\boldsymbol{u}_j = \boldsymbol{D}_j(\boldsymbol{y}_{\mathrm{obs}})$, define the response ray $\boldsymbol{y}_j(r) = \boldsymbol{z}_j + \sigma r \boldsymbol{u}_j$, $r \geq 0$, and the event-preserving radial region

$$\mathcal{Z} = \{r \geq 0 : \widehat{\ell}(\boldsymbol{y}_j(r)) = \ell_0,\ M(\boldsymbol{y}_j(r)) = M_0\}. \tag{14}$$

The event fixes the unordered selected set, not its internal ranking. Because v is fixed, it also preserves the target identity. For each $a \in M_0$ and $b \in [m] \setminus M_0$, membership requires

$$\boldsymbol{y}_j(r)^\top (A^{\lambda_0}_{a,[m]} - A^{\lambda_0}_{b,[m]}) \boldsymbol{y}_j(r) \geq 0, \tag{15}$$

and, for every $\ell \in \{1, \ldots, 9\} \setminus \{\ell_0\}$, it requires

$$\boldsymbol{y}_j(r)^\top (C_\ell - C_{\ell_0}) \boldsymbol{y}_j(r) \geq 0. \tag{16}$$

Equality is admitted only when the relevant index tie rule favors the observed choice. For either comparison, let $F \in \mathbb{R}^{n\times n}$ denote the corresponding symmetric matrix difference in (15) or (16). The comparison is quadratic in r:

$$\sigma^2(\boldsymbol{u}_j^\top F \boldsymbol{u}_j)r^2 + 2\sigma(\boldsymbol{u}_j^\top F \boldsymbol{z}_j)r + \boldsymbol{z}_j^\top F \boldsymbol{z}_j \geq 0. \tag{17}$$

Intersecting their solution sets gives a finite union of intervals, up to endpoints of zero continuous probability. Both identically tied comparisons and their tie rules must be retained. Importantly, the CV matrices in (16) act on the original response space; CV need not depend only on its projection into the full-data basis space.

Write $f_{\chi_{d_j}}$ for the density of the chi distribution with d_j degrees of freedom and $\mathbf{1}\{\cdot\}$ for the indicator function. Conditioning on $\ell_0, M_0, \boldsymbol{z}_j, \boldsymbol{u}_j$ below means conditioning on $\widehat{\ell}(\boldsymbol{Y}) = \ell_0$, $M(\boldsymbol{Y}) = M_0$, $\mathcal{Q}_{\boldsymbol{Y}} = \boldsymbol{z}_j$, and $\boldsymbol{D}_j(\boldsymbol{Y}) = \boldsymbol{u}_j$. For positive selection probability, the conditional density of $R_j(\boldsymbol{Y})$ under the applicable null H_0 is

$$f(r \mid X, \omega, v, \ell_0, M_0, \boldsymbol{z}_j, \boldsymbol{u}_j; H_0) = \frac{f_{\chi_{d_j}}(r)\mathbf{1}\{r \in \mathcal{Z}\}}{\int_{\mathcal{Z}} f_{\chi_{d_j}}(t)\,dt}. \tag{18}$$

With $r_{\text{obs}} = R_j(\boldsymbol{y}_{\text{obs}})$, compute the selective upper-tail probability

$$p_{\text{selective},j,U} = \frac{\int_{\mathcal{Z}\cap[r_{\text{obs}},\infty)} f_{\chi_{d_j}}(t)\,dt}{\int_{\mathcal{Z}} f_{\chi_{d_j}}(t)\,dt}. \tag{19}$$

We evaluate interval masses using chi distribution functions and logarithmic arithmetic. For a zero-rank target, set $p_{\text{selective},j,U} = 1$. For positive-rank targets, assume $r_{\text{obs}} > 0$ and positive finite selection mass; these conditions hold almost surely under the nondegenerate conditional model.

In exact arithmetic, (18)-(19) yield a uniform conditional p-value under the corresponding fixed-label null, for nondegenerate conditioning values. In particular, for every $\boldsymbol{\mu}$ satisfying that null and every $\alpha \in (0,1)$,

$$\begin{aligned}\mathbb{P}_{\boldsymbol{\mu}}(p_{\text{selective},j,U} \leq \alpha \mid X, \omega, v, \widehat{\ell}(\boldsymbol{Y}) = \ell_0,\ M(\boldsymbol{Y}) = M_0,\\ \mathcal{Q}_{\boldsymbol{Y}} = \boldsymbol{z}_j,\ \boldsymbol{D}_j(\boldsymbol{Y}) = \boldsymbol{u}_j) \leq \alpha.\end{aligned} \tag{20}$$

For the zero-rank convention, rejection probability is zero. The Gaussian assumption, known scale, specified test space, and faithful event computation are essential to this statement; floating-point implementation requires separate numerical validation.

For a prespecified significance level $\alpha \in (0,1)$, reject when $p_{\text{selective},j,U} < \alpha$. The procedure tests one uniformly selected feature for a prespecified target definition. The guarantee is for that individual test. Reporting multiple discoveries or choosing a target definition after examining its p-value requires an additional error-control argument.